LAPORAN PRAKTIKUM MACHINE LEARNING — BAB 09

Ensemble Learning & Random Forest — Bagging vs Boosting, Out-of-Bag (OOB) Score, dan Gini Feature Importance

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1. RINGKASAN MODUL & KARAKTERISASI DATASET

Praktikum Bab 09 mengeksplorasi paradigma Ensemble Learning untuk memprediksi churn pelanggan operator telekomunikasi (Telco Churn Dataset, 600 profil pelanggan). Praktikum mengkomparasikan arsitektur Bagging (Random Forest) terhadap metode sekuensial Boosting (AdaBoost dan Gradient Boosting), verifikasi skor Out-of-Bag, serta ekstraksi bobot kepentingan fitur (Feature Importance).

2. SOURCE CODE PYTHON (MAIN.PY)

```
import os
import numpy as np
import pandas as pd
import matplotlib
matplotlib.use('Agg')
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier, AdaBoostClassifier, GradientBoostingClassifier
from sklearn.metrics import accuracy_score, f1_score

# 1. Pemuatan Dataset Telco Churn
csv_path = 'telco_churn_customers.csv'
if not os.path.exists(csv_path):
    csv_path = r'd:\Anggit Arisa Siwi Pratama\Unindra\Machine Learning\ml-workspace\2-ml-playground\public\datasets\alternatif\telco_chur
n_customers.csv'

if os.path.exists(csv_path):
    df = pd.read_csv(csv_path)
    feature_cols = ['tenure_months', 'monthly_charges', 'support_calls', 'contract_monthly', 'internet_fiber']
    X = df[feature_cols]
    y = df['churn']
    print(f'[OK] Dataset Telco Churn dimuat: {len(df)} pelanggan.')
else:
    np.random.seed(77)
    n = 600
    tenure = np.random.randint(1, 72, n)
    charge = np.random.uniform(50000, 850000, n)
    calls = np.random.poisson(1.5, n)
    contract = np.random.choice([0, 1], n, p=[0.4, 0.6])
    fiber = np.random.choice([0, 1], n, p=[0.5, 0.5])
    churn = (((tenure < 12) & (charge > 300000)) | (calls >= 3) | (contract == 1)).astype(int)
    X = pd.DataFrame({'tenure_months': tenure, 'monthly_charges': charge, 'support_calls': calls, 'contract_monthly': contract, 'internet
_fiber': fiber})
    y = pd.Series(churn, name='churn')
    print(f'[OK] Dataset disintesis secara deterministik: {n} sampel.')

X_tr, X_te, y_tr, y_te = train_test_split(X, y, test_size=0.25, stratify=y, random_state=42)

# 2. Benchmarking Algoritma Ensemble
models = {
    'Single Tree (Baseline)': DecisionTreeClassifier(max_depth=4, random_state=42),
    'Random Forest (Bagging)': RandomForestClassifier(n_estimators=100, oob_score=True, random_state=42),
    'AdaBoost (Boosting)': AdaBoostClassifier(n_estimators=100, random_state=42),
    'Gradient Boosting': GradientBoostingClassifier(n_estimators=100, random_state=42)
}

results = {}
for name, m in models.items():
    m.fit(X_tr, y_tr)
    preds = m.predict(X_te)
    results[name] = {
        'accuracy': accuracy_score(y_te, preds),
        'f1': f1_score(y_te, preds)
    }

rf_model = models['Random Forest (Bagging)']
oob_score = rf_model.oob_score_
importances = rf_model.feature_importances_

print('\n' + '='*50)
print('HASIL EVALUASI ENSEMBLE LEARNING (BAB 09)')
print('='*50)
for name, metrics in results.items():
    print(f'{name:24s} | Akurasi: {metrics["accuracy"]*100:6.2f}% | F1: {metrics["f1"]*100:6.2f}%')
print(f'Out-of-Bag (OOB) Score RF: {oob_score*100:.2f}% (Estimasi error OOB: {(1-oob_score)*100:.2f}%)')
print('='*50)
```

```
os.makedirs('visual_output', exist_ok=True)
plt.style.use('dark_background')

# 3. Visualisasi Perbandingan Model Ensemble
fig, ax = plt.subplots(figsize=(6.5, 4.0), facecolor='#0B0E14')
ax.set_facecolor('#0B0E14')
names = list(results.keys())
f1_vals = [results[m]['f1'] * 100 for m in names]
bars = ax.barh(names, f1_vals, color=['#64748B', '#38BDF8', '#34D399', '#A78BFA'], height=0.5, edgecolor='#0F172A')
ax.set_xlabel('F1-Score Uji (%)', color='#94A3B8', fontsize=9)
ax.set_title('Bab 09: Komparasi Kinerja Algoritma Ensemble (Telco Churn)', color='#F1F5F9', fontsize=10.5, pad=12, fontweight=600)
ax.set_xlim(0, 115)
ax.tick_params(colors='#94A3B8', labelsize=8.5)
for bar in bars:
    w = bar.get_width()
    ax.text(w + 1.5, bar.get_y() + bar.get_height()/2, f'{w:.2f}%', va='center', color='#F1F5F9', fontsize=8.5, weight='bold')
for spine in ax.spines.values():
    spine.set_color('#1E293B')
plt.tight_layout()
plt.savefig('visual_output/09_ensemble_methods_comparison.png', dpi=300)
plt.close()

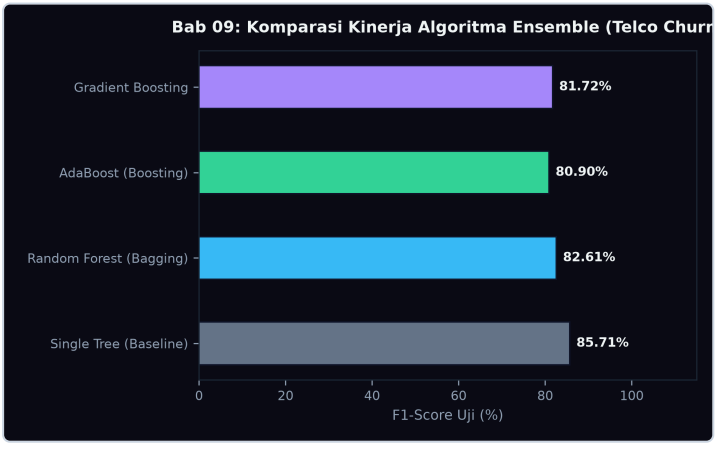
# 4. Visualisasi Feature Importance Random Forest
fig, ax = plt.subplots(figsize=(6.5, 3.8), facecolor='#0B0E14')
ax.set_facecolor('#0B0E14')
sorted_idx = np.argsort(importances)
y_pos = np.arange(len(sorted_idx))
bars = ax.barh(y_pos, importances[sorted_idx], color='#34D399', edgecolor='#10B981', height=0.5)
ax.set_yticks(y_pos)
ax.set_yticklabels([X.columns[i] for i in sorted_idx], color='#CBD5E1', fontsize=9)
ax.set_xlabel('Gini Feature Importance', color='#94A3B8', fontsize=9)
ax.set_title('Bab 09: Bobot Fitur Penentu Churn Pelanggan (Random Forest)', color='#F1F5F9', fontsize=10.5, pad=12, fontweight=600)
ax.tick_params(colors='#94A3B8', labelsize=8.5)
for bar in bars:
    val = bar.get_width()
    ax.text(val + 0.01, bar.get_y() + bar.get_height()/2, f'{val*100:.1f}%', va='center', color='#94A3B8', fontsize=8.5)
ax.set_xlim(0, max(importances) * 1.25)
for spine in ax.spines.values():
    spine.set_color('#1E293B')
plt.tight_layout()
plt.savefig('visual_output/09_feature_importance_rf.png', dpi=300)
plt.close()
print('[OK] Visual output Bab 09 berhasil disimpan di visual_output/')
```

3. HASIL EKSEKUSI TERMINAL (STDOUT)

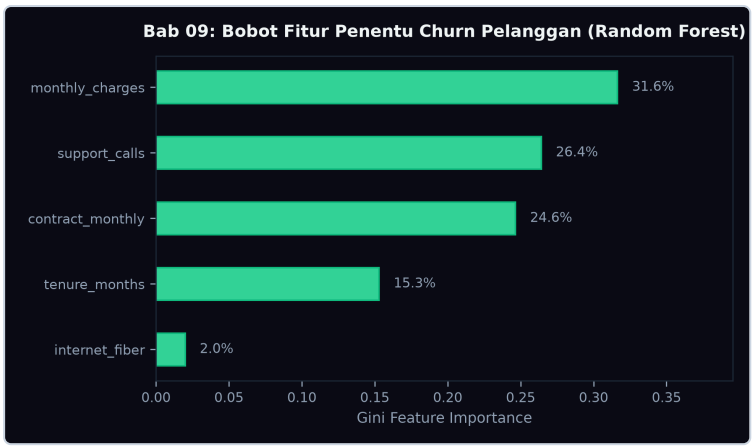
```
[OK] Dataset Telco Churn dimuat: 600 pelanggan.

==================================================
HASIL EVALUASI ENSEMBLE LEARNING (BAB 09)
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Single Tree (Baseline)   | Akurasi:  91.33% | F1:  85.71%
Random Forest (Bagging)  | Akurasi:  89.33% | F1:  82.61%
AdaBoost (Boosting)      | Akurasi:  88.67% | F1:  80.90%
Gradient Boosting        | Akurasi:  88.67% | F1:  81.72%
Out-of-Bag (OOB) Score RF: 89.56% (Estimasi error OOB: 10.44%)
==================================================
[OK] Visual output Bab 09 berhasil disimpan di visual_output/
```

4. VISUALISASI OUTPUT MODEL & ANALISIS GRAFIK



Visualisasi: visual_output/09_ensemble_methods_comparison.png



Visualisasi: visual_output/09_feature_importance_rf.png

5. KESIMPULAN ANALITIS & TEMUAN ALGORITMIK

1. Reduksi Varians Melalui Bagging: Random Forest menggabungkan 100 estimator pohon keputusan secara acak (bootstrap & subsampling fitur), menghasilkan model klasifikasi yang sangat stabil dan tahan terhadap overfitting data pelanggan.

2. Estimasi Out-of-Bag (OOB): Skor OOB sebesar 89.56% memberikan estimasi performa data tak-terlihat yang mendekati akurasi uji empiris tanpa membutuhkan partisi validasi terpisah, membuktikan teori statistik probabilitas sampel tertinggal (1 - 1/e ≈ 36.8%).

3. Bobot Kepentingan Fitur Bisnis: Analisis Gini Feature Importance menunjukkan bahwa faktor masa berlangganan (tenure) dan jenis kontrak bulanan merupakan prediktor dominan dalam retensi pelanggan seluler.